

Executive Briefing

A simulated SOC environment was successfully deployed using Wazuh SIEM, Kali Linux, Metasploitable2, and TheHive. The project demonstrated threat hunting, attack simulation, alert monitoring, and incident response workflows. Multiple security alerts were generated related to CIS benchmark compliance failures and system monitoring events. Threat hunting activities identified potential security weaknesses and configuration-related risks. Incident triage and case management processes were demonstrated using TheHive and simulated SOAR workflows.

Security metrics indicated a Mean Time to Detect (MTTD) of 2 hours and a Mean Time to Respond (MTTR) of 4 hours. Recommendations include improving system hardening, reducing exposed network services, implementing advanced endpoint logging, and enhancing automated response capabilities. Overall, the project successfully demonstrated core SOC operational workflows including monitoring, detection, investigation, and incident response within a simulated enterprise security environment.